

WKT2, axis order, datum grids

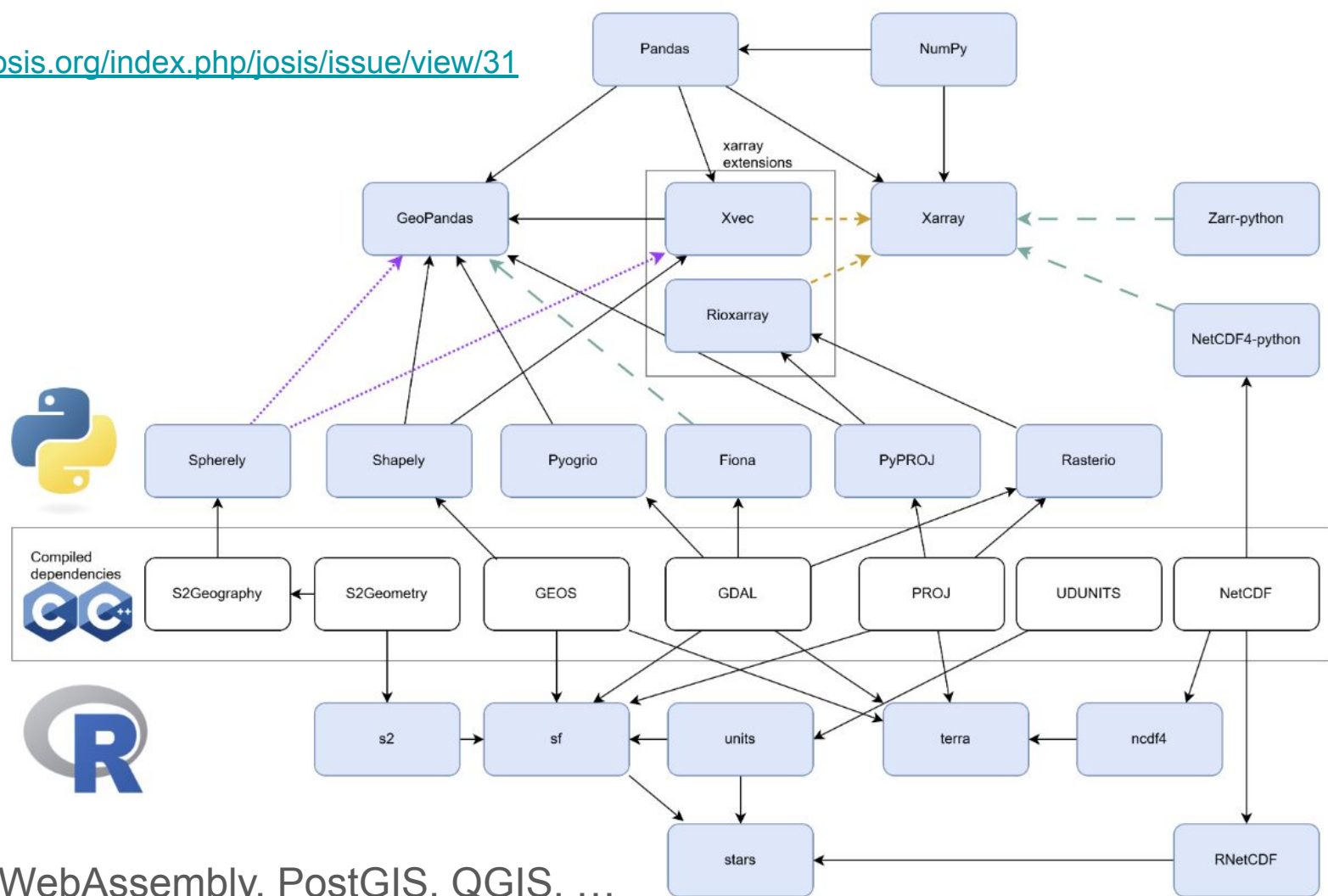
Reference Systems, Jun 24, 2026

What is WKT2?

- WKT: well-known-text
- A text (“ascii file”) standard for (i) geometries: points/lines/polygons, (ii) coordinate reference systems
- “WKT-2”: informal name of the WKT standard for CRS
- An OGC standard
- https://en.wikipedia.org/wiki/Well-known_text_representation_of_coordinate_reference_systems
- <https://www.ogc.org/standards/wkt-crs/>
- Editor: Roger Lott - sounds familiar?
- ProjJSON also exists... but not (yet?) as a formal standard

What is EPSG?

- An organisation, now [LOGP](https://logp.org/)
- A database with “registered” coordinate reference system
- This database is part of PROJ
- What is PROJ? See <http://proj.org/>
- Where did PROJ come from? From Proj.4, Gerald Evenden, USGS, dating back to the 70s, since 2019 an OSGEO project with a PSC
- 2016: Thomas Knudsen adds coordinate transformation
- 2018: support for OGC WKT, late-binding coordinate transformation model (no longer a WGS84 “hub”)



Axis order

- What does “X - Y” mean?
- For matrix A, $A[i,j]$ refers to row i, column j: $i=y, j=x$
- “There is no natural ordering”
- Putting North up is (only) a convention

Axis order

- South Africa has a family of South Orientated Transverse Mercator coordinate systems in which the axes are x=westing, y=southing, e.g. EPSG:2046
- “Standard” geographic coordinates (geodetic coordinates, long/lat degrees) are not “standard”:
- The “most used” EPSG:4326, specifies: latitude, longitude
- OGC simple features specifies coordinates as x, y
- OGC:CRS84 specifies: longitude, latitude
- Most (?) dataset “out there”, specify EPSG:4326 **BUT** maintain longitude,latitude order
- However, a standards-compliant web feature server (WFS) when asked for data in EPSG:4326 SHOULD give latitude,longitude coordinates
- Summarizing: a mess

cdn.proj.org

- What is a [cdn](#)?
- Contains open data licensed datum grids
- Datum transformation is hard, and approximate!
- One can download the datum grids, and look up values
- ... or let PROJ do this:
 - It will read the pixels / section needed, efficiently (from a COG)
 - It will cache the region(s) read in a local user directory, for subsequent use
 - ... if you let it do so!